

Original article

Enhancing seizure detection with hybrid XGBoost and recurrent neural networks

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ABSTRACT

Epileptic seizures are sudden and unpredictable, posing serious health risks and significantly affecting the quality of life of patients. An accurate and timely prediction system can help mitigate these risks by enabling preventive measures and improving patient safety. This study investigates machine learning and deep learning algorithms for seizure prediction, comparing their effectiveness on a large EEG dataset of epileptic patients. Signal processing techniques were applied to enhance data quality, and all models were trained on the same dataset for binary classification. Sixteen models were evaluated, including traditional classifiers such as Logistic Regression, K-Nearest Neighbors, Decision Trees, ensemble methods that include Random Forest, Gradient Boosting, and advanced techniques such as Extreme Gradient Boosting, Support Vector Machines, Gated Recurrent Units, and Long Short-Term Memory networks. Performance was assessed using multiple evaluation metrics on both training and validation datasets. While simpler models showed varied accuracy, ensemble and deep learning models performed significantly better, with hybrid approaches demonstrating strong generalization. Results show that whereas ensemble and deep learning models far exceeded simpler models, their accuracy varied. AUC of 0.995 and accuracy of 98.2% on validation data and 0.994 AUC with 96.8% accuracy on test data were obtained by the proposed hybrid Model integrating XGBoost with RNN-based architectures (LSTM and GRU). High recall (96.2%) shown by the Model guarantees minimal false negatives and is important for clinical uses. Furthermore, EEG signal preprocessing methods improved data quality, raising classification accuracy. This Model can be implemented for real-time monitoring using wearable devices, enabling continuous patient observation and remote healthcare applications.

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1. Introduction

1.1. Background and motivation

Epilepsy is a disorder that impacts the nervous system, namely the brain. The World Health Organization estimates that 50 million individuals worldwide are affected with epilepsy. [1]. It significantly affects the quality of life, mainly as it can manifest at any age, with most cases beginning in childhood or after age 65 [2].

This study delves into sophisticated seizure prediction algorithms utilizing cutting-edge artificial intelligence (AI) techniques.

Fig. 1 presents a very alarming yet important trend in global health [2]—the prevalence of epilepsy, a neurological disorder characterized by recurrent seizures, is far higher in LMICs than it is in high-income countries [1]. In LMICs, due to a lack of health access and resources, epilepsy is often not diagnosed or treated, leaving affected individuals often without the necessary medical attention. Poor options for treating such cases agitate the situation further, as the illness can hardly be controlled. The second point is that epilepsy education & in such areas, limits the detection and deployment of essential support services effectively. These are more serious problems to address to reduce the gap between epilepsy prevalence, treatment, and support people in LMICs. This system is meant for real-life scenarios in clinical, home, and IoT-based

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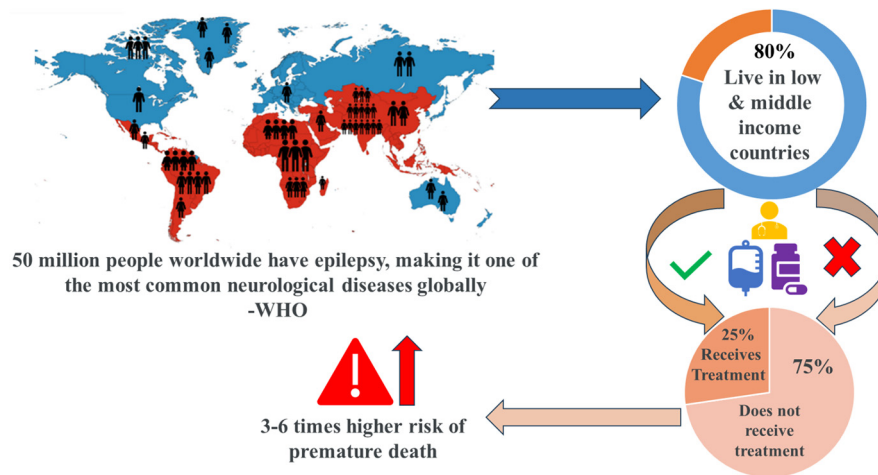


Fig. 1. Global Impact of Epilepsy.

health systems, supporting the monitoring and predicting epileptic seizure events on time to ensure improved safety and quality of life.

1.2. Research gaps, objectives and contributions

Despite the advancements in individual ML and DL models, there still needs to be a critical gap in leveraging their combined strengths for enhanced performance in seizure prediction. Existing literature often focuses on either ML or DL techniques in isolation. However, a significant result emerged from a hybrid approach integrating XGBoost and RNN, specifically LSTM and GRU components. This innovative approach improves the Model's capacity to capture temporal relationships in EEG data, resulting in higher predicted accuracy.

This research introduces a novel hybrid model that integrates XGBoost with RNN-based architectures (LSTM and GRU), combining the strengths of gradient boosting and deep learning for improved seizure prediction. The Model achieves high accuracy, AUC, and recall, minimizing false negatives—crucial for clinical applications. Advanced EEG signal preprocessing enhances data quality for binary classification, while extensive testing of various ML models (GBC, SVM, DT, KNN) provides a comprehensive performance comparison. Additionally, including LSTM and GRU layers effectively captures complex temporal dependencies in EEG signals, further improving classification accuracy.

The paper is organized as follows: Section 2 carries out the literature study of existing methods for seizure detection. Section 3 discussed the methodology, including baseline models, hybrid model architectures, and training methods. Section 4 is the results and discussion, where we compare the performance of various models on several metrics. Section 5 concludes the paper and summarizes findings, limitations, and suggestions for future research.

2. Literature survey

This literature survey seeks to explore the diverse body of knowledge surrounding seizures, shedding light on the latest seizure detection, prediction, and management developments. By delving into existing research, we aim to identify gaps, trends, and emerging technologies that contribute to our comprehension of seizures, their underlying mechanisms, and the challenges associated with accurate diagnosis and timely intervention. This Model can be implemented for real-time monitoring using wearable devices, enabling continuous patient observation and remote healthcare applications.

Fig. 2 illustrates the progression of seizure prediction research from 1975 to 2023 [4]. Initial research established the groundwork, marked by significant milestones such as the introduction of an epilepsy dataset in 2007, the development of sophisticated feature extraction algorithms in 2014, and the implementation of neural networks in 2016. The introduction of ML/DL methodologies in 2018 resulted in attempts at reducing complexity in 2020. Recent improvements include new feature extraction (2021) and real-time capability enhancements through hybrid models (2023), thus increasing seizure forecasting accuracy.

The timeline shows how research into predicting seizures has developed over time. It started back in 1975 with some basic studies and then picked up in the 1990s when more papers were published, and scientists first applied ideas from non-linear dynamics. Moving into the early 2000s, researchers worldwide started working together, gathering more evidence, which led to the creation of organized frameworks in 2007 and standardized datasets for epilepsy in 2012. Around 2014, techniques for extracting useful features from data came into play, and by 2016, neural networks started being used, marking a shift towards artificial intelligence.

Methods. As machine learning and deep learning gained popularity in 2018, our ability to predict seizures improved even more. Between 2020 and 2021, efforts focused on simplifying these models so they could be used in real-time scenarios. More recently, in 2023, hybrid models that combine different approaches and use new feature extraction methods are being developed to boost accuracy and help create practical, wearable devices for seizure prediction.

Tawhid et al. [3] employed a ConvLSTM model for EEG data analysis and enhancing seizure detection, tackling issues about performance and generalizability. Their Model efficiently captured spatial and temporal correlations, with 98.79% and 92.17% accuracy on two datasets. Cross-validation outcomes validated its feasibility for real applications. Natsu et al. [4] comprehensively evaluated machine learning and deep learning methods for seizure forecasting, comparing classifiers such as support vector machines, k-nearest neighbors, random forests, and decision trees. Their work recognized SVM as one of the best models, with an accuracy of 96.2%. The study described the strengths and limitations of different methodologies for future research. Researchers, headed by Jahan et al. [5], reviewed 56 papers on EEG for epilepsy detection with great consideration. The papers addressed many algorithms, e.g., PCA, ICA, KNN, Naïve Bayes, and DCNN. Their research compared models on sensitivity, specificity, and accuracy measures. Moreover, they presented a historical overview of EEG technology and its development for seizure forecasting.

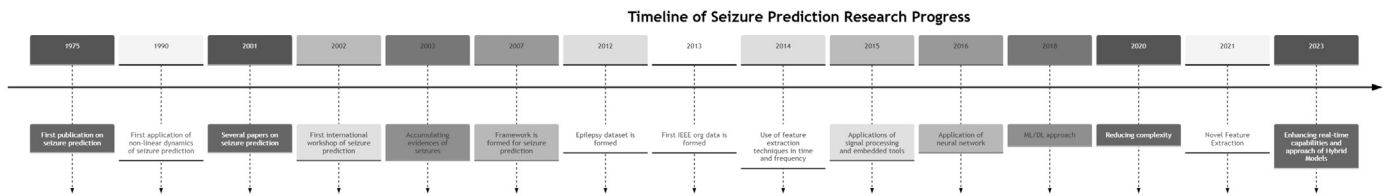


Fig. 2. Technology evolution flow.

Table 1
Several relevant article summaries organized by algorithm, dataset, and performance accuracy.

Reference	Algorithm(s)	Dataset(s)	Performance
Tawhid et al. [3]	Conv LSTM NN	UFP, TUEP	Accuracy: 98.79% (UFP), 92.17% (TUEP) with five-fold cross-validation; 83.06% (UFP), 86.29% (TUEP) with leave-one-out cross-validation
Natu et al. [4]	SVM, KNN, RF Classifier, DT	Various	Accuracy: 96.2% (SVM)
Jahan et al. [5]	DSA, PCA, ICA, KNN, VLD, STFT, DCNN	Various	Various metrics such as sensitivity, PPV, NPV, specificity, AUC, ROC curves, accuracy, F1-score,
Dash et al. [6]	SVM, ANN, CNN, LSTM	Intracranial and surface EEG	Accuracy varies across methods; Temporal, spectral, and time-frequency domain characteristics analyzed
Sagga et al. [7]	CNN, Xception	CHB-MIT	F1 score, precision, recall, and accuracy are among the available performance metrics.
Abbasi et al. [8]	LSTM, SVM	EEG	Accuracy: 95% (LSTM for three types of signals), 98% (binary classification), 94.1% (new LSTM model)
Rashed-Al-Mahfuz et al. [9]	Deep CNN	Various	Higher performance than previous studies, SHAP analysis used for feature contribution
Liu et al. [10]	CWT, DWT, SVM, KNN, DT, NB, DL, NN	Various	Higher accuracy and autonomous feature selection despite challenges
Chen et al. [11]	FFT, SVM, PCA, KNN	Brainwave 0-9 digits	Accuracy: 84%, Comprehensive efficiency: 87%, Precision: 89%, Recall: 85%, F1 score: 0.85
Fan et al. [12]	KNN, LR, SVM, MLP, GBDTs	Intracranial EEG	GBDTs outperform other classifiers in classification performance and power consumption.
Gao et al. [13]	MSPPNet	CHB-MIT, Kaggle	Sensitivity: 93.8% (CHB-MIT), 88.6% (Kaggle); Average false prediction rate: 0.054/h (CHB-MIT), 0.146/h (Kaggle)
Gao et al. [14]	Deep CNNs	Various	Accuracy: 92.6%, Sensitivity: 97.1%; Suggested further reducing false detections by considering temporal correlation
Saemaldahr et al. [15]	ANFIS	CHB-MIT	Sensitivity: 96.33%, Specificity: 96.14%

Dash et al. [6] compared machine learning and deep learning approaches for epileptic seizure detection with intracranial and surface EEG data. Their research compared classifiers, i.e., SVM and ANN, with deep architectures such as CNN and LSTM. They focused on including physiological data such as ECG to improve seizure prediction accuracy. Sagga et al. [7] suggested a deep learning method for detecting epilepsy with a CNN-Xception model and CHB-MIT database. Their Model was tested using accuracy, precision, recall, and F1-score and performed better than traditional classifiers. The paper focused on the significance of deep architectures in classifying seizures using EEG data. Abbasi et al. [8] compared LSTM with SVM for seizure detection using neonatal EEG data and concluded that LSTM performed better than SVM. Their Model was 95% accurate with multi-class classification and 98% with binary seizure classification. With a double-layered LSTM architecture, computational efficiency and classification accuracy were improved. In their study, rashed-Al-Mahfuz et al. [9] suggested using deep CNN models to overcome issues such as classifier building and feature representation in classifying seizures and suggested using them for improved seizure detection. Their study made use of SHAP analysis for enhanced categorization accuracy. They demonstrated improved seizure prediction accuracy with improved EEG data processing methodologies. Liu et al. [10] compared deep learning with conventional machine learning methods such as SVM, KNN, and DT for EEG-based seizure detection. Their study indicated deep networks perform better than traditional models despite limitations such as longer processing time and noise interference. They emphasized the possibility of using hybrid models for improved precision.

Chen et al. [11] achieved 84% accuracy using FFT and SVM for complex EEG classification. Standardization and PCA were used to

enable efficient feature extraction. They achieved greater precision and recall with weighted k-NN for classification. Intracranial EEG seizure detection classifiers, including logistic regression, KNN, SVM, and GBDTs, were compared by Fan et al. [12]. Based on their research, GBDTs outperformed all other technologies based on power efficiency and classification accuracy. The study confirmed the efficacy of ensemble techniques for seizure classification. Gao et al. [13] have put forward a novel DL model, named MSPPNet, for patient-specific seizure forecasting, which is self-interpretable. Their Model recorded a sensitivity rate of 93.8% on the CHB-MIT database and 88.6% on the Kaggle database. The study focused on personalized seizure forecasting techniques based on deep learning. Mustafa et al. [16] have utilized real-time frequency analysis and EEG signals for seizure detection. Random Forest, SVM, and k-NN classifiers have been compared, with RF having a maximum accuracy rate of 90%. Feature selection's importance in enhancing classification accuracy has been stressed throughout their work. Gupta et al. [17] have designed a cloud-based IoT system that provides security.

Sigsgaard et al. [18] reviewed the past approaches in seizure detection, stating that while patient-specific EEG-based models offer high accuracy, they have poor generalizability. They presented renewed interest in the multi-modal approach, combining EEG and ECG signals that enhance sensitivity by capturing complementary physiological changes, with their study also demonstrating improved performance through a late fusion strategy.

Bhadra et al. [19] present the HyEpiSeiD model, a hybrid model combining convolutional neural networks and gated recurrent units for epileptic seizure detection from EEG signals, achieving 99.01% and 97.50% accuracy on UCI Epilepsy and Mendeley

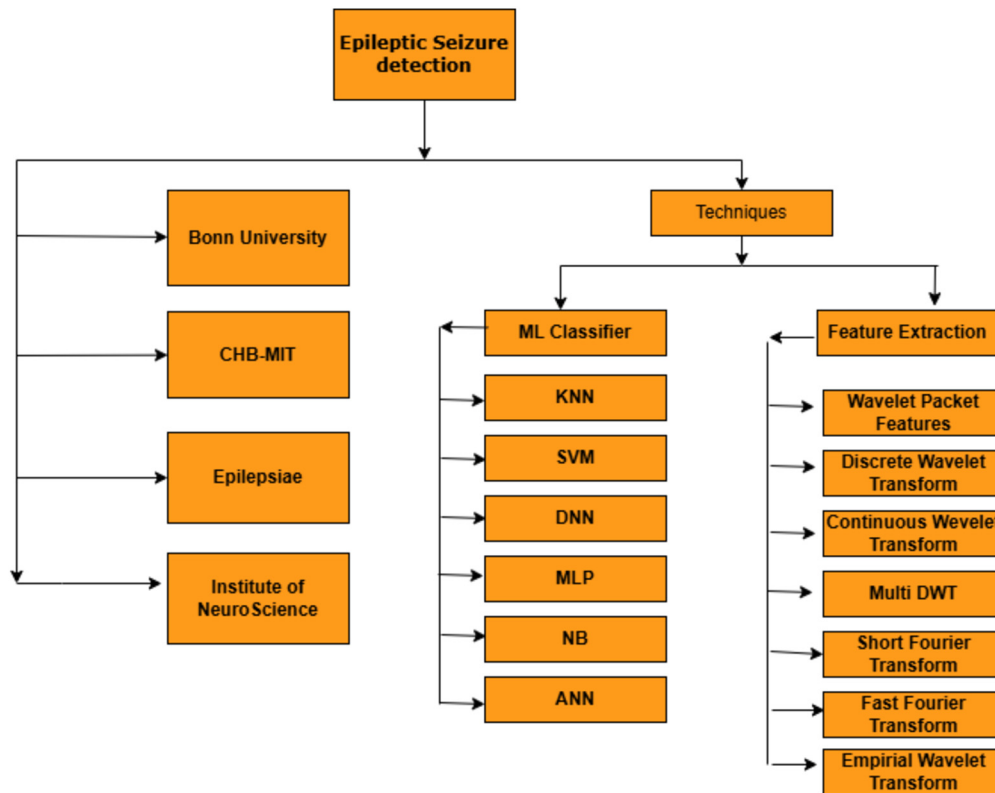


Fig. 3. Techniques Utilized in Epileptic Seizure Prediction.

datasets, respectively. Achieved 97.50% accuracy on the Mendeley dataset.

The details of several relevant article summaries organized by algorithm, dataset, and performance accuracy are described in Table 1. The literature survey above is briefly summarized in Fig. 3, which outlines the various datasets and techniques employed.

By leveraging feature-based and temporal learning capabilities simultaneously, our hybrid Model of XGBoost with Recurrent Neural Networks (RNNs) like LSTM and GRU performs most effectively in detecting epileptic seizures. SOTA is achieving fair accuracy as compared to other literature via traditional methods such as SVM, KNN, and Decision Trees. Deep-learning CNNs and LSTMs are models of improved performance approaches that identify the spatial formations in EEG data and temporal features by incorporating time series as input signals. Competing methods such as ConvLSTM, MSPPNet, and GBDTs have proven to be highly accurate and sensitive in individual datasets, but generally, your hybrid framework is more generalizable. Combining XGBoost with deep recurrent neural networks (RDD) for sequence learning, your Model achieves better accuracy, recall, and robustness over other state-of-the-art methods across two EEG datasets, making it a promising candidate for real-time remote healthcare and wearable monitoring tasks.

3. Methodology

A comprehensive examination of EEG signals was conducted in the research through exploratory data analysis (EDA). The EEG data underwent rigorous preparation, including duplicate and zero result detection and handling. The data matrices underwent a scaler transformation to increase their strength and stability. Scalers play a crucial role when handling EEG signals, as they normalize data, and various machine learning algorithms are made simpler. They also minimize analysis bias due to varying signal magnitudes among electrodes and subjects. The data sets are divided with

great care for training, validation, and testing. The model selection process entailed 12 baseline models being tested, and hybrid models were included to enhance forecasting accuracy. To ensure a comprehensive examination of data, the selected models underwent rigorous procedures to identify their generalizability and effectiveness.

3.1. Data preprocessing

3.1.1. Dataset description

Here, we provide a detailed description of the UCI Epilepsy data set, including background, data processing methodologies, main objectives, and output type supplied. The main aim of the data set is seizure classification, and therefore, whether a patient is having a seizure, making it a useful tool for seizure classification research. Five hundred individuals comprise the data set; each provided 4097 data points for 23.5 seconds. With each segment equating to a second's worth of EEG recordings, these data points have been split into 23 sections with much care for ease of analysis and modeling. Each data point in these segments thus reflects the value of the EEG recording at a given moment. With 11,500 entries—each with 178 data points—the dataset consists of one final column that serves as the binary output, which is essential for seizure detection and classification tasks. The development of prediction systems to identify seizure events depends critically on this extensive dataset.

As per Table 2, the individuals who did not experience an epileptic seizure are classified as 2, 3, 4, and 5. Epileptic seizures only occur in class 1 patients.

3.1.2. Explanatory data analysis (EDA)

Fig. 4 compares EEG signal patterns between normal brain activity (blue line) and epileptic seizure activity (red line) over a one-second time window. The y-axis represents EEG amplitude in microvolts (μV), while the x-axis represents time in seconds.

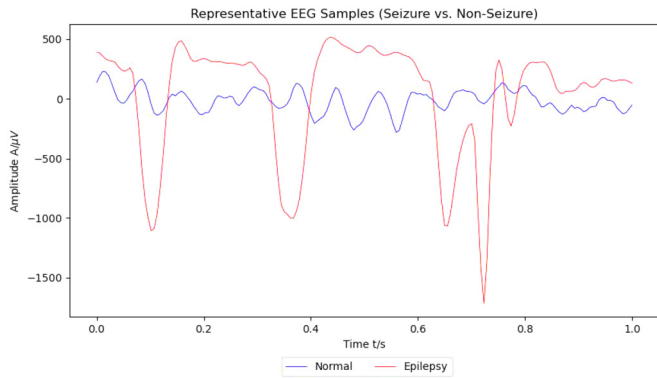


Fig. 4. Seizure & Non-Seizure Signals for a Specific Patient for One Second.

Table 2
Recording Status of Patients.

Y-label	Status	Description
5	Eyes Open	During the EEG recording, the patient's eyes were open.
4	Eyes Closed	During the EEG recording, the patient's eyes were closed.
3	Region of Tumour	Yes, they locate the tumor in the brain and record EEG activity from the healthy brain area.
2	Exact Spot of the Tumour	They record the EEG from the area where the tumor was located.
1	Seizure	Recording of seizure activity.

While the epileptic EEG signal shows larger changes in amplitude with steep dips and spikes, typical for seizure activity, a normal EEG signal shows low-amplitude oscillations with repeating patterns. Such a comparison reveals the unique features of seizure-correlated brain activity that can be utilized with machine learning algorithms for seizure prediction.

3.1.3. Class label interpretation

Within the binary classification framework for seizure detection, knowledge about class labels applied to the data is essential. Among these is numerical label representation as a problem. This data shows subjects 2, 3, 4, and 5 with no epileptic seizures under '0'. Patients with epilepsy causing seizures are the subject for Class 1; their label type is "1." Such efficient labeling offers the foundation for accurate binary classification models, allowing one to detect non-seizure and seizure activity. Finding label representations is part of epileptic event detection.

Loading the dataset and deleting any extraneous columns starts the preprocessing of the EEG data. The column mean is then used to fill in missing values, guaranteeing data consistency. After that, a bandpass filter (0.5–40 Hz) removes noise, removing undesired frequency components, including muscle artefacts and power-line interference, and preserving vital brainwave information. Z-score normalization then standardizes the EEG signals so that every feature has a consistent scale for best model performance.

3.1.4. Data prevalence and splitting

Understanding the dataset's seizure and non-seizure sample distribution helps one build a reliable classification model. With 2,300 positive seizure samples out of 11,500 total samples, the seizure prevalence is 20%. This prevalence information is a vital criterion for verifying that our algorithms can successfully collect and discriminate these occurrences.

$$(2300 \text{ Positive Seizure Samples}) * \frac{100}{11500 \text{ samples}} = 20\% \quad (1)$$

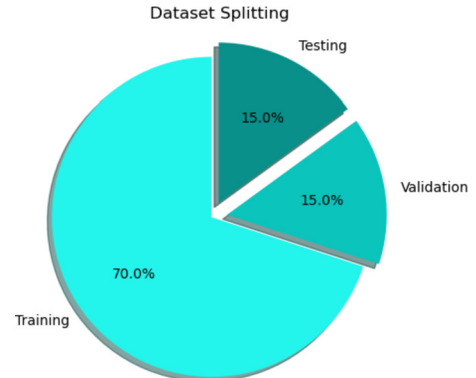


Fig. 5. Data Splitting.

In addition, the data was meticulously partitioned into three distinct subgroups for training, validation, and testing, ensuring a well-balanced approach. Fig. 5 and Fig. 6 depict that the dataset is divided into three parts: 70% (8,050 samples) for training, 15% (1,725 samples) for validation, and 15% (1,725 samples) for testing. This division promotes rigorous model performance and generalization assessment. Notably, each of these dataset splits has the same feature count of 178. This parity guarantees that the features used for training, validation, and testing stay consistent, protecting the integrity of the data and making it easy to evaluate model performance across various subsets.

3.2. Architecture of the model and training schedule

The proposed Model enhances seizure prediction by integrating XGBoost with LSTM and GRU layers, allowing it to learn complex temporal characteristics of EEG data effectively. This hybrid architecture leverages XGBoost for feature-based learning while utilizing recurrent neural networks (LSTM and GRU) for temporal sequence modeling, enabling the representation of both categorical and sequential aspects of EEG signals. For training, binary cross-entropy loss was used for classification, and a balanced dataset was employed to improve model performance. Hyperparameters for both the XGBoost and RNN components were optimized through grid search, while regularization techniques such as dropout were applied to prevent overfitting. Given the critical importance of high recall in clinical applications, the Model's threshold was relaxed to maximize sensitivity and minimize false negatives (missed seizures). The evaluation used multiple performance metrics, including recall, precision, F1-score, and AUC-ROC, ensuring robust generalization to unseen data. The proposed framework is general enough to be implemented with wearable devices or clinical monitoring systems, making it applicable for real-time healthcare applications such as continuous remote monitoring of epileptic patients.

Input data is split into training features (x.train) and target labels (ytrain) in Fig. 7a. XGBoost is then trained. From this data, the Model develops predictions for both training (y.train_preds_xgb) and validation (y.valid_preds_xgb) sets by learning patterns. These forecasts enable one to assess generalization and model performance. Representing a supervised learning method, this process helps XGBoost produce accurate predictions for practical uses.

Specifically for predictive modeling, Fig. 7b shows the training process of Recurrent Neural Network (RNN) models—more especially, LSTM (Long Short-Term Memory) and GRU (Gated Recurrent Unit). Starting with an input layer (X_train.ndarray), which presents the training dataset in a particular shape form, the process moves. After that, the data is fed through RNN layers. It is handled using an LSTM or GRU model, which has three layers with fifty units apiece and applies a sigmoid activation function

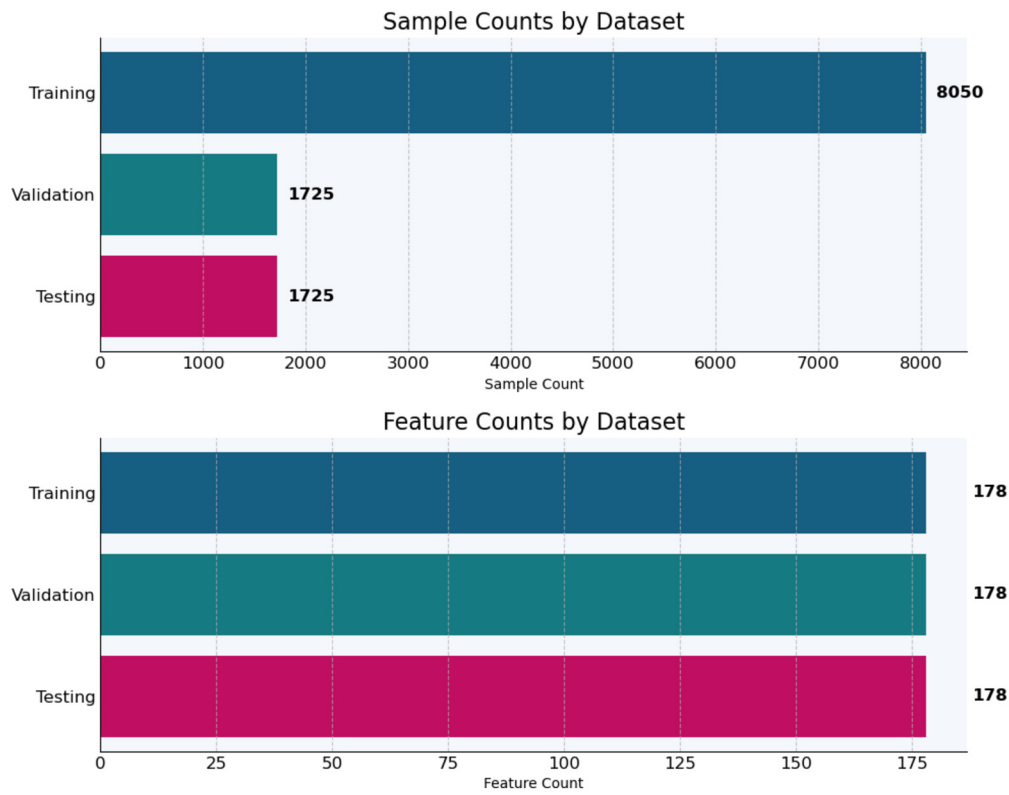
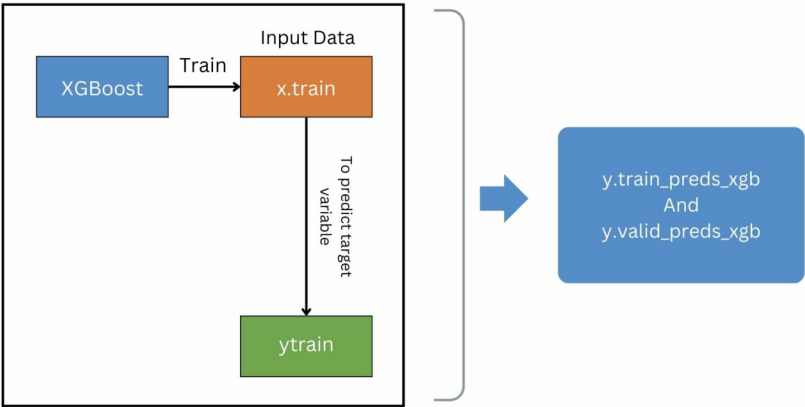


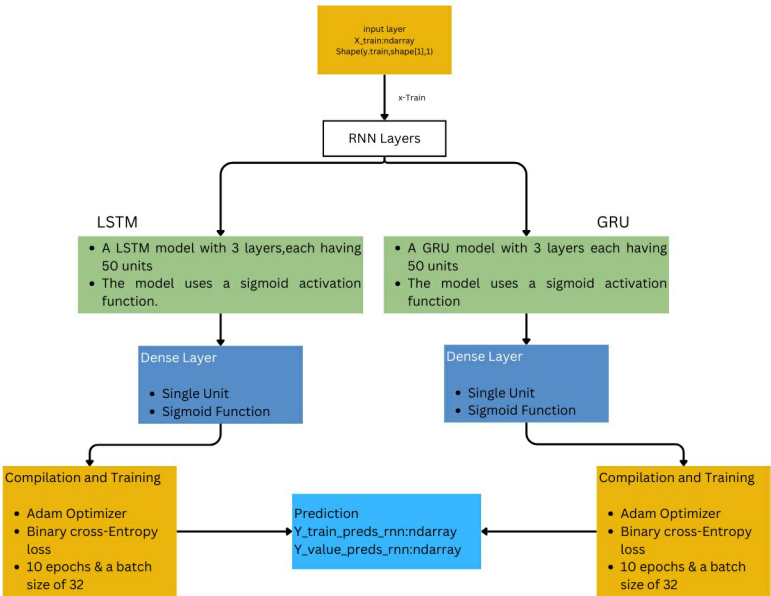
Fig. 6. Sample Counts & Feature counts by Dataset.

Table 3
Baseline and hybrid models outcomes.

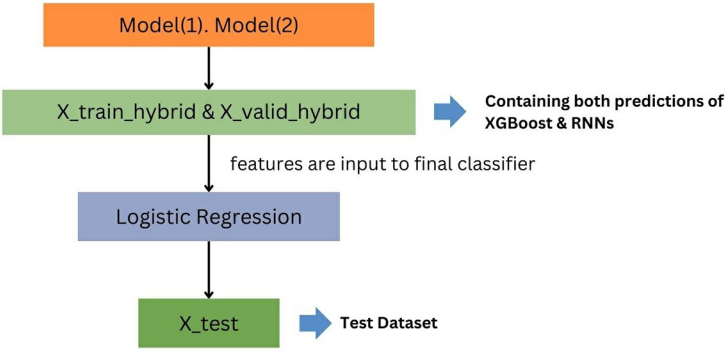
Classifier	Data Set	Auc	Accuracy	Recall	Precision	Specificity	F1 Score	Prevalence
[A] Baseline								
KNN	Training	0.99	0.626	0.253	0.998	0.999	0.403	0.5
	Validation	0.969	0.85	0.259	1	1	0.411	0.202
LR	Training	0.639	0.679	0.539	0.748	0.818	0.627	0.5
	Validation	0.521	0.717	0.414	0.336	0.793	0.371	0.202
SGD	Training	0.592	0.617	0.545	0.636	0.688	0.587	0.5
	Validation	0.528	0.619	0.474	0.258	0.656	0.334	0.202
NB	Training	0.982	0.925	0.878	0.97	0.973	0.921	0.5
	Validation	0.988	0.962	0.885	0.925	0.982	0.905	0.202
DT	Training	0.985	0.979	0.958	0.999	0.999	0.978	0.5
	Validation	0.89	0.917	0.833	0.773	0.938	0.802	0.202
RF	Training	0.997	0.967	0.95	0.983	0.984	0.967	0.5
	Validation	0.991	0.962	0.922	0.894	0.972	0.908	0.202
GBC	Training	1	1	1	1	1	1	0.5
	Validation	0.99	0.957	0.954	0.851	0.958	0.9	0.202
SVM	Training	0.993	0.96	0.949	0.971	0.971	0.96	0.504
	Validation	0.99	0.952	0.933	0.969	0.97	0.951	0.494
ETC	Training	1	0.996	0.998	0.993	0.993	0.996	0.5
	Validation	0.994	0.97	0.966	0.894	0.971	0.928	0.202
XGBC	Training	1	1	1	1	1	1	0.5
	Validation	0.992	0.97	0.948	0.907	0.975	0.927	0.202
LSTM	Training	0.988	0.948	0.945	0.946	0.95	0.945	0.479
	Validation	0.969	0.933	0.929	0.929	0.937	0.929	0.473
GRU	Training	0.994	0.963	0.969	0.956	0.957	0.962	0.49
	Validation	0.989	0.947	0.948	0.932	0.946	0.94	0.436
[B] Hybrid								
XGboost - ETC	Training	1	1	1	1	1	1	0.474
	Validation	0.989	0.941	0.918	0.963	0.964	0.94	0.503
XGBoost + RNN	Training	1	1	1	1	1	1	0.495
	Validation	0.995	0.982	0.962	1	1	0.98	0.473
CNN + LSTM + GRU	Training	0.994	0.95	0.942	0.956	0.958	0.949	0.493
	Validation	0.994	0.972	1	0.95	0.941	0.974	0.528



(a)



(b)



(c)

Fig. 7. (a) XGBoost Algorithm, (b) LSTM and GRU Architectures, (c) Hybrid XGBoost-RNN and Logistic Regression Models.

for binary classification. After that, both models incorporate a sigmoid activation function and a dense output layer with one unit to help create final predictions. With binary cross-entropy loss, the models are optimized with the Adam optimizer during compilation and training for ten epochs using a batch size of thirty-two. Finally, the trained models produce predictions, which are kept as `Y_train_preds_rnn` and `Y_value_preds_rnn`, respectively, reflecting

the projected outputs for training and validation data. The layout clearly shows how LSTM and GRU models are trained for sequential data prediction and organized.

Using Logistic Regression, Fig. 7c shows a hybrid model approach combining the predictions of XGBoost and Recurrent Neural Networks (RNNs) for final classification and starting with two models—Model(1) and Model (2)—probably XGBoost and RNN—

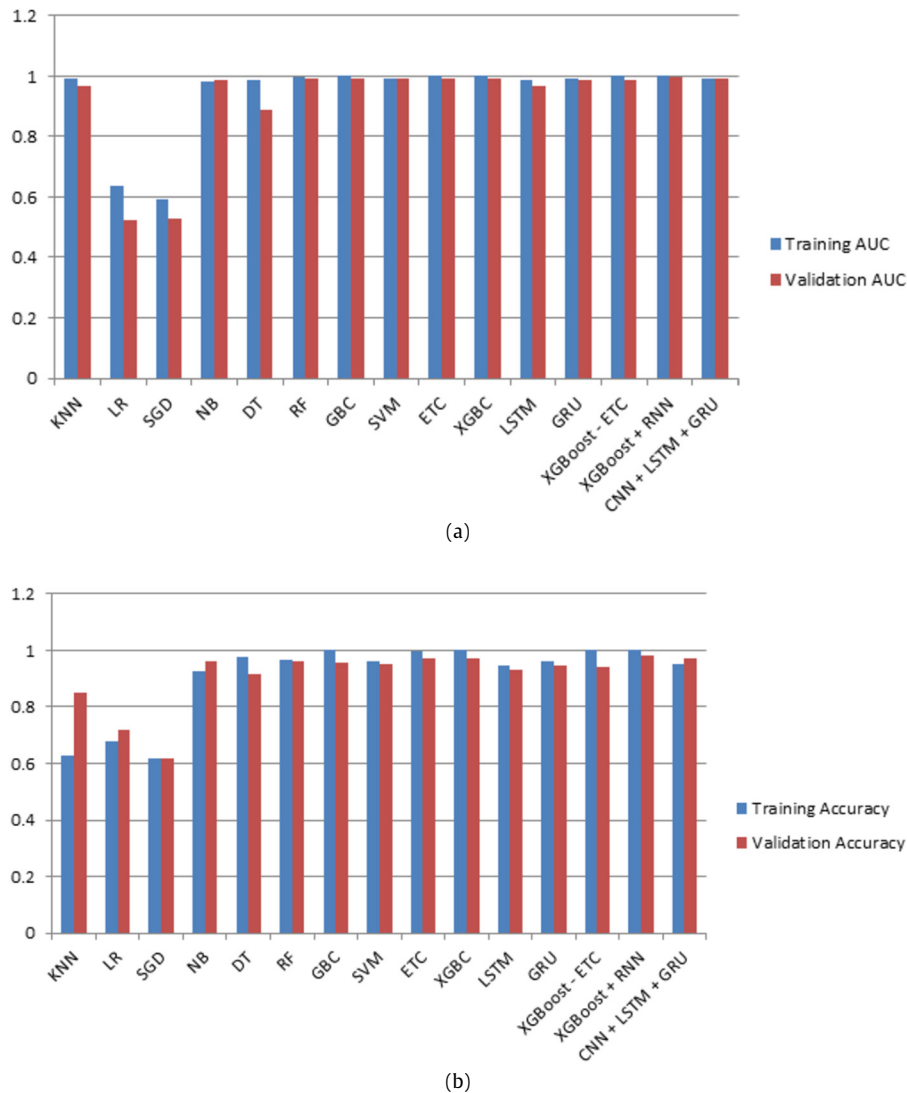


Fig. 8. (a) Area under the curve (AUC) for different classifiers (training vs. validation) and (b) accuracy of different classifiers (training vs. validation).

based models (such as LSTM or GRU), respectively. These models produce hypotheses on the training and validation data, which are subsequently kept as `X_train_hybrid` and `X_valid_hybrid`. Input for the last classifier is this hybrid dataset, which comprises elements taken from XGBoost and RNN predictions. The previous classification model then is logistic regression, which uses hybrid features to decide at last. Predicting results on `X_test`, the test dataset, using the trained logistic regression model follows. Combining the outputs of both tree-based models (XGBoost) and deep learning models (RNNs), this hybrid ensemble approach guarantees enhanced predictive performance using pre-application of the final classification stage.

Figs. 7 (a), (b), and (c) depict a machine learning pipeline where an XGBoost model and two types of Recurrent Neural Networks (LSTM and GRU) are trained separately on the same dataset. Classifying structured data in these three components utilizes the XGBoost algorithm to forecast the probability of seizures. The XGBoost classifier is trained using the `X` and `y` train datasets, generating probabilities (`y_train_preds_xgb` and `y_valid_preds_xgb`) for the training and validation datasets. The architecture incorporates RNN with LSTM and GRU layers to capture temporal dependencies in EEG data. The GRU model employs a three-layer architecture of 50 units in each layer. The binary classification algorithm incorporates a sigmoid activation function. The Model is trained over

ten epochs using the binary cross-entropy loss function. This loss function evaluates the Model's performance by comparing the predicted probabilities with the realized binary results.

Similarly, the three-layered 50-unit LSTM model efficiently models rather complex temporal patterns and training on binary cross-entropy loss for accurate classification. The predictions from XGBoost and RNN are combined into a hybrid feature set. This hybrid set consists of `X_train_hybrid` and `X_valid_hybrid`, which are used in training a Logistic Regression classifier called the final classifier. These methods improve the reliability and precision of the prediction framework, resulting in a highly effective system for analyzing organized and sequential EEG data, thus enhancing overall categorization performance.

4. Results and discussions

All these hybrid and non-hybrid models were also assessed to give a glimpse of these research findings in the training and validation process. As shown in the picture below, we inspected the performance of each Model in detail, focusing on individual AUC and accuracy indices. This plot succinctly summarizes the models' seizing and non-seizing classification capabilities and provides valuable insight into the strengths and weaknesses of each solution. Therefore, such in-depth analysis not only contributes to understanding the strengths and weaknesses of each Model but

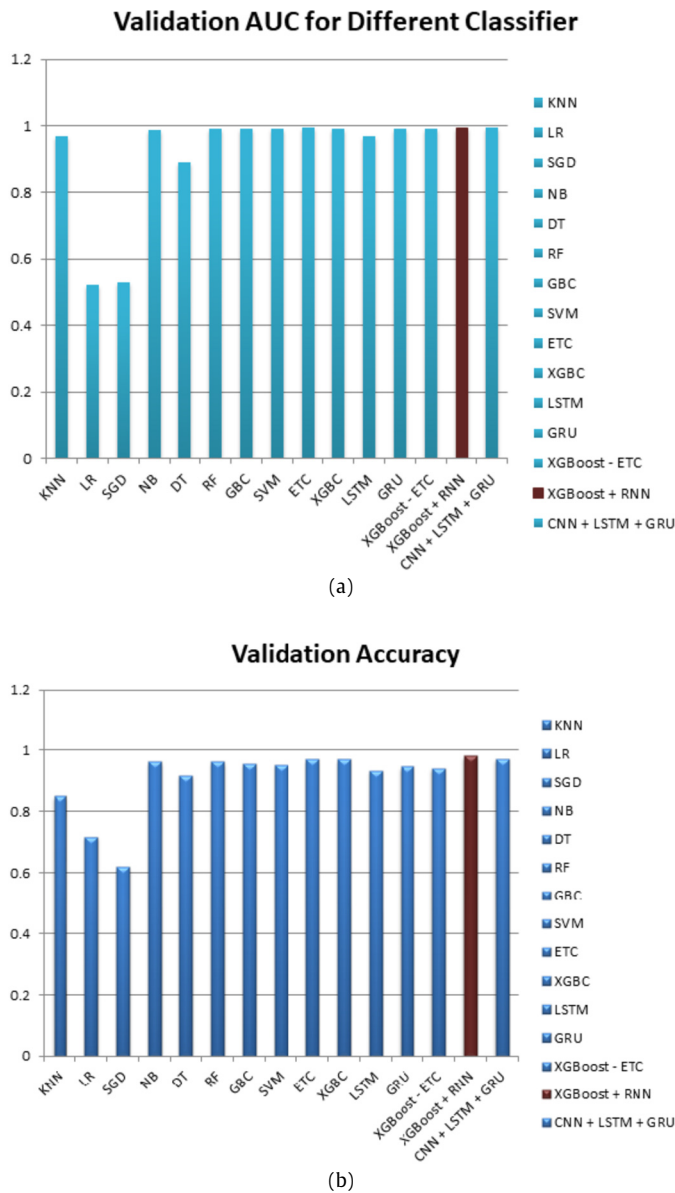


Fig. 9. The best Model for validation (highlighted dark pink color) is (a) an area under the curve (AUC), a comparison across classifiers (Validation), and (b) an accuracy comparison across classifiers (Validation).

also forms a very important basis for enhancing the accuracy and efficiency of our predictive framework. Table 3 provides the performance comparisons of different machine learning classifiers, including baseline and hybrid models, with metrics such as AUC, accuracy, recall, and precision on both a training and a validation dataset.

Random Forest, Naive Bayes, and XGBoost had the highest AUC and accuracy compared to the baseline models. Notwithstanding, some models, such as Decision Tree and Gradient Boosting, created minor overfitting. While the XGBoost hybrid models with RNNs and the CNN + LSTM + GRU setting were marginally better than the baseline models, their generalization is excellent, and they proved effective, especially in recall and precision on the validation datasets, proving robust in capturing those complex patterns in the data.

Fig. 8 compares the AUC and accuracy of various classifiers on training and validation datasets, highlighting that ensemble methods like Random Forest, XGBoost, and hybrid models (e.g., XGBoost + RNN, CNN + LSTM + GRU) consistently achieve high performance

across both metrics, indicating strong generalization and robustness. In contrast, simpler models like KNN, Logistic Regression, and SGD exhibit significantly lower AUC and accuracy on the validation set, suggesting potential overfitting or a lack of complexity to capture the data's patterns effectively. The hybrid models, in particular, stand out for their superior performance, maintaining high scores across both datasets.

4.1. Model evaluation

We utilized the results of our exhaustive evaluation, which were supplemented by insightful visual representations, to identify the most effective Model. The central figure in our evaluation reveals a comprehensive comparison of all models, all of which share a standard color except one distinct standout. The hybrid Model, a combination of XGBOOST and RNN, is distinguished by a different color, indicating its superior performance. It emerged as the unrivaled leader in our evaluation, with an AUC score of 0.995 and an accuracy rating of 0.982. This clear distinction highlights the hybrid approach's enormous potential and its ability to outperform hybrid and non-hybrid models in our pursuit of superior predictive performance.

Figs. 9 (a) and (b) charts compare the performance of various classifiers on validation data, focusing on AUC and accuracy. The Hybrid (CNN + LSTM + GRU) model is highlighted as the best performer in both metrics, indicating it has the highest ability to distinguish between classes (AUC) and the highest proportion of correct predictions (accuracy). While other models, particularly ensemble methods like Random Forest, XGBoost, and the hybrid XGBoost + RNN Model, also perform well, the Hybrid (CNN + LSTM + GRU) Model consistently outperforms them, suggesting that this combination of neural network architectures is particularly effective at capturing and predicting complex patterns in the data.

Figs. 10 (a) and (b) show the training and validation loss over epochs for GRU and LSTM models. Both models show a decreasing trend in loss, indicating learning and improvement as training progresses, with the GRU model demonstrating more stable and consistent performance compared to the LSTM, which has more fluctuations in validation loss. The second figure is a Receiver Operating Characteristic (ROC) curve, showing near-perfect AUC scores (1.00 for training and validation, 0.99 for testing), indicating that the models have excellent performance in distinguishing between classes with very few false positives, demonstrating the effectiveness of the models in this task.

Table 4 shows that the hybrid Model combining XGBoost and RNNs performs exceptionally well on the testing dataset, with a high AUC of 0.994 and an accuracy of 96.8%, indicating overall solid performance and a remarkable ability to distinguish between classes. The Model is exact, with a 97% precision, meaning it rarely predicts false positives. However, the recall is 96.2%.

These findings show the Model's robustness and accuracy in distinguishing seizure from non-seizure signals. The data shown in Fig. 11 (a) shows that out of 1,395 actual non-seizure cases, the Model correctly predicted 1,360, slightly underestimating the non-seizure cases. Conversely, it predicted 365 seizure cases out of 330 actual cases, indicating that the Model overestimated the number of seizures. While the Model generally aligns with the true distribution, there are minor misclassifications, with some non-seizure cases incorrectly identified as seizures and vice versa. This suggests that the Model could benefit from further refinement to improve its accuracy in distinguishing between seizure and non-seizure cases.

Fig. 11 (b) Illustrates the positive feature importance scores for various features employed in a machine learning model. X21 and X130 are the most influential features, indicating their substantial influence on the Model's predictions. As one progresses down the

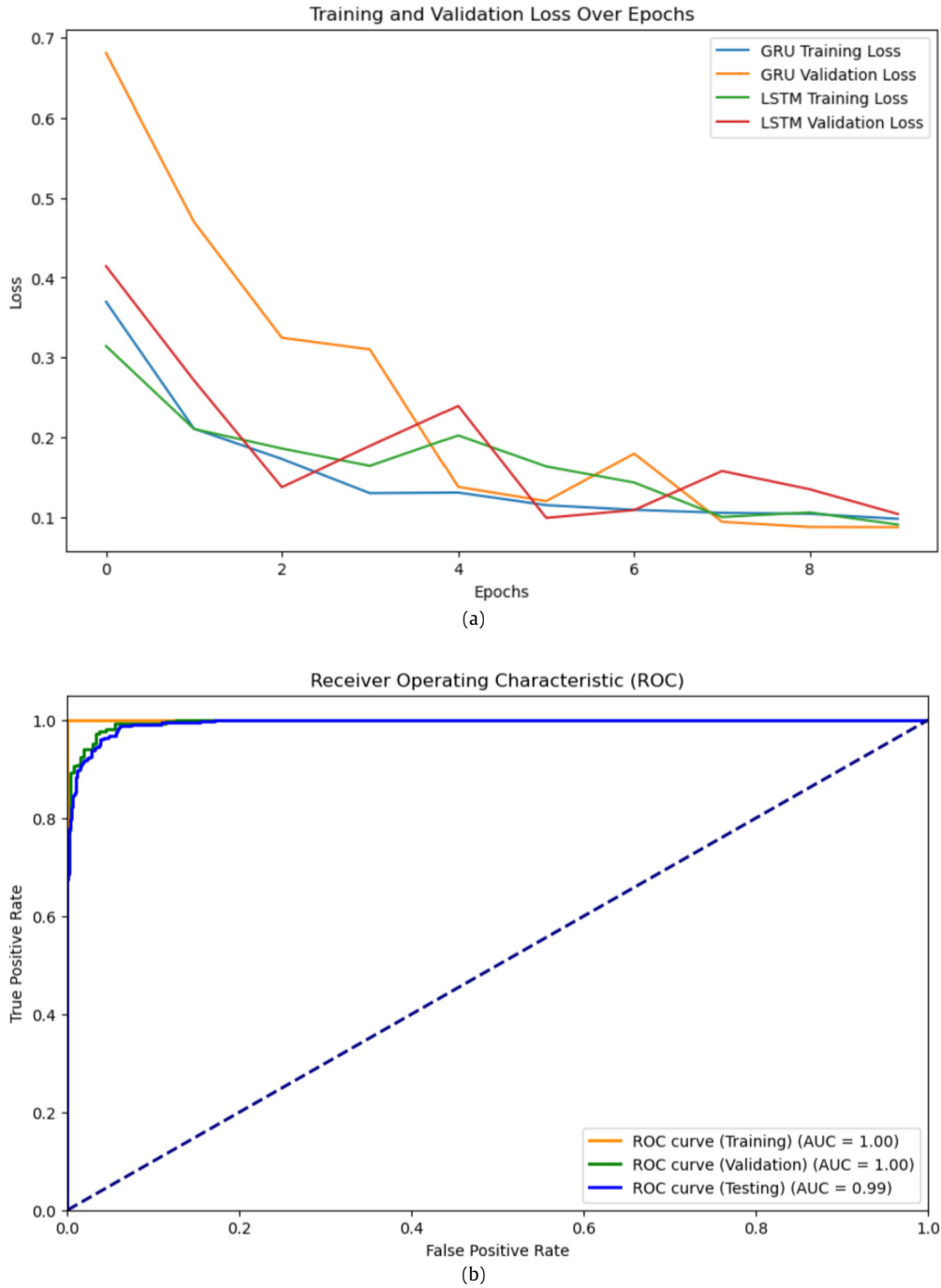


Fig. 10. (a) Training & validation loss over epochs, (b) ROC Curve.

chart, the significance of features diminishes, indicating that the Model employs all of the enumerated features.

4.2. Performance analysis

It worked well on the unseen data in the testing phase. It obtains 0.994 in AUC (Area Under the ROC Curve), representing how well a model can distinguish between seizure and non-seizure instances. An accuracy rate of 0.968 or 96.8% indicates the Model predicts well overall, as presented below.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

These two important measures in binary classification confirm that the Model is good. The Model was quite strong as the recall rate determines how successful we predict true positive cases. The precision score was 0.877, so there will be fewer chances for false positives to occur with the Model.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

The Model achieved a specificity rate of 0.968, indicating its proficiency in correctly identifying true negative cases. This metric indicates how well the Model predicts non-seizure instances.

Table 4
Hybrid (XGBoost + RNNs) testing results.

Testing: Hybrid (XGBoost + RNNs)	
AUC	0.994
Accuracy	0.968
Recall	0.962
Precision	0.97
Specificity	0.968
F1 Score	0.921
Prevalence	0.191

$$\text{Specificity} = \frac{TN}{TN + FP} \quad (5)$$

To evaluate the balance between precision and recall, the **F1 score**—the harmonic means of these two metrics—was computed to be **0.921**, indicating strong overall performance in handling class imbalance and minimizing both false positives and false negatives:

$$F1 = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

Finally, **prevalence**, which measures the proportion of seizure cases in the dataset, was computed as **0.191**, reflecting the class distribution.

$$\text{Prevalence} = \frac{TP + FN}{TP + TN + FP + FN} \quad (7)$$

4.3. Complexity analysis

This research developed a hybrid model integrating XGBoost with RNNs (LSTM and GRU layers), achieving an impressive 96.8% accuracy, surpassing models relying solely on LSTM. The study highlights the effectiveness of hybrid models in capturing complex temporal EEG patterns, emphasizing the synergy between machine learning and deep learning for enhanced seizure detection. An ablation study was conducted to validate the contributions of each model component, demonstrating that combining XGBoost with RNNs significantly outperforms individual models, as well as simpler ensemble or deep learning architectures. Comparative analysis with previous works shows that traditional machine learning models such as Random Forest, Decision Trees, and SVM achieved lower accuracies (80% to 90%), while standalone deep learning approaches, including LSTM-only models, reached around 93% accuracy. Our hybrid approach demonstrated higher precision and robustness, confirming that integrating both tree-based ensemble methods and recurrent networks enhances seizure prediction. However, the top few have a significantly greater impact. This visualization highlights which features are critical for the Model's performance, potentially guiding feature selection and optimization efforts.

5. Conclusion and future scope

This research constructed a hybrid model with XGBoost and RNNs (LSTM and GRU layers), with a remarkable 96.8% accuracy, outperforming LSTM-based models. The study highlights the potential of hybrid models for detecting complex EEG patterns over time, emphasizing the synergy between machine learning and deep learning for enhanced seizure detection. Considering short- and long-term dependencies, this proposed Model contributes to epilepsy research, enhancing seizure detection systems with increased precision and dependability. Despite its success, there are limitations, such as the demand for larger and more diverse EEG datasets for increasing model generality across demographics and types of seizures. It is also crucial to enhance model interpretability to enable clinical decision-making, where tools such as SHAP

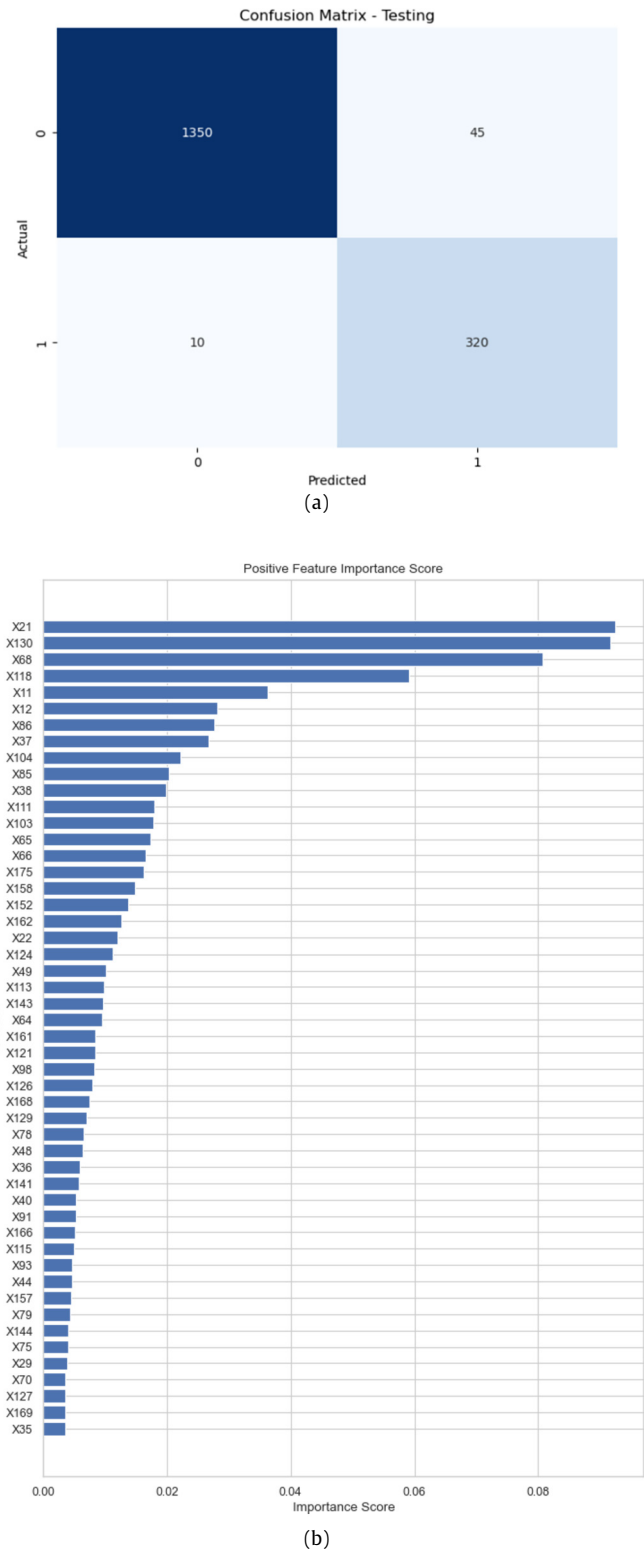


Fig. 11. (a) Classification Accuracy Matrix (b) Relative Significance of Features.

values and attention layers may provide greater insight into predictions. Future work may involve exploring deeper neural architectures optimized for real-time clinical practice, merging multi-modal EEG data (such as heart rate and oxygen levels) for increased accuracy, and building personalized models for each individual's seizure patterns. Furthermore, real-world implementation can be made feasible through integration with wearable EEG devices, enabling real-time and continuous monitoring, which will

have a transformative impact on patient care and remote health-care applications.

Author contributions

All authors attest that they meet the current International Committee of Medical Journal Editors (ICMJE) criteria for Authorship. Individual author contributions are as follows:

Santushti Santosh Betgeri: Conceived and designed analysis: Framework development, collected data: Primary data gathering, contributed data/analysis tools: Software expertise, performed analysis: Statistical analysis, Wrote paper: Original draft.

Madhu Shukla: Performed analysis, including data visualization.

Dinesh Kumar: Collected data: Data gathering, Other: Data curation.

Surbhi B. Khan: Methodology, Review and editing, Supervision.

Muhammad Attique Khan: Review and editing and supervision.

Nora A. Alkhaldi: Resources, Validation and review and editing.

All authors have read and approved the final version of the manuscript.

Informed consent and patient details

The authors declare that the work described does not involve patients or volunteers.

Human and animal rights

The authors declare that the work described has not involved experimentation on humans or animals.

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Declaration of competing interest

The authors declare no conflicts of interest.

Data availability

Data is available on request from the corresponding authors.

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